

LYRA THE LYRE

THIS PROMINENT CONSTELLATION IN THE NORTHERN SKY CONTAINS THE FIFTH-BRIGHTEST STAR VISIBLE FROM EARTH, VEGA, ALONG WITH SEVERAL INTERESTING DOUBLE STARS AND A FAMOUS PLANETARY NEBULA.

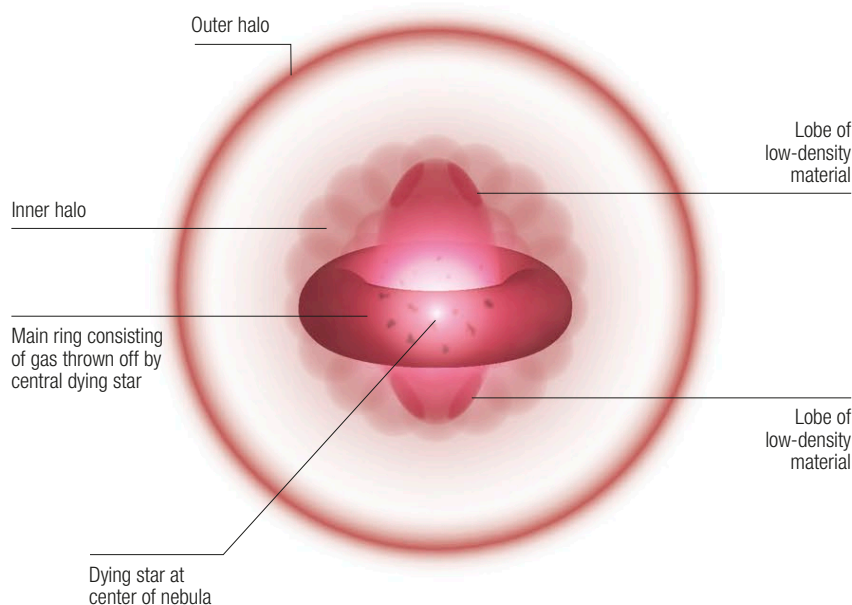
The constellation Lyra is said to represent the lyre, or harp, played by the legendary Greek musician Orpheus. However, Arab astronomers visualized it as an eagle or vulture, and the name of its brightest star, Vega, comes from an Arabic phrase meaning “swooping eagle (or vulture).”

Near the brilliant Vega lies Epsilon Lyrae, a celebrated quadruple star about 160 light-years away. Telescopes show that each star in this “double” is in fact a close pair—hence its popular name, the Double Double. Another famous double is Beta Lyrae. In this case, the brighter component is also an eclipsing binary (see p.43), which varies between magnitudes 3.3 and 4.4 every 12.9 days.

Delta Lyrae is an unrelated pair of red and blue-white stars, of 4th and 6th magnitudes, which is easily divided with binoculars. Zeta Lyrae, another pair of 4th and 6th magnitudes, can be separated with binoculars or small telescopes. Between Beta and Gamma Lyrae lies the Ring Nebula, M57, a beautiful planetary nebula shaped like a smoke ring (see pp.122–23).

▽ Structure of M57 (Ring Nebula)

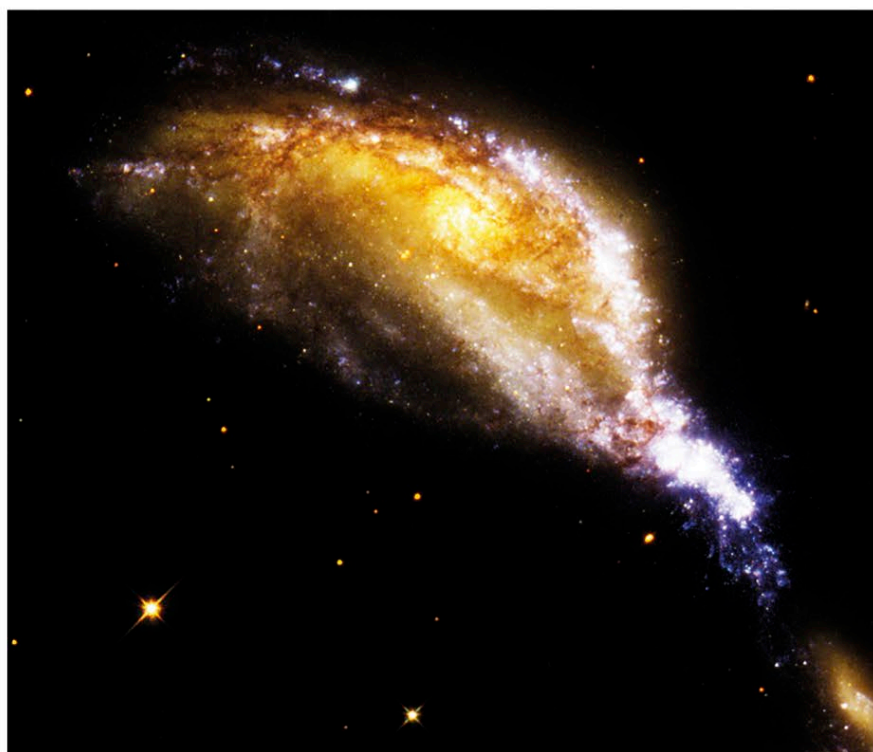
From our viewpoint on Earth, the planetary nebula M57 looks like a smoke ring around a central star. Seen side-on, though, it would look more like the diagram above. A doughnut-shaped ring of gas with denser knots in it is expanding away from the central star's equator, while fainter lobes of thinner gas extend from the star's poles (top and bottom).



Epsilon Lyrae, the **Double Double**, is a remarkable family of four stars, all linked by gravity

▷ NGC 6745

A collision between two galaxies has produced this strangely shaped object, which resembles a bird's head. The main part of the “head,” seen here in this Hubble Space Telescope image, is a spiral galaxy. It has been highly distorted by its encounter with a smaller elliptical galaxy, which is just visible in the bottom right corner, at the end of the “beak.” The blue-white patches at the top and right of the spiral are areas of star formation triggered by the collision.



△ M56

This globular cluster is a ball of ancient stars just over 30,000 light-years from the Sun. It requires a fair-sized telescope to be seen well. Studies of the chemical composition and age of the stars in the cluster suggest that it was once part of an older dwarf galaxy that subsequently merged with the Milky Way.